SURABHI BHADAURIA
CURRICULUM VITAE

Aerospace Engineering, Embry-Riddle Aeronautical University (ERAU), Daytona Beach, FL, 32114
+1(765) 775-9123, bhadaurs@erau.edu

RESEARCH INTERESTS

Cislunar space domain awareness, space situational awareness, orbital mechanics and dynamics, trajectory design, observations, detection and tracking

EDUCATION

Ph.D., Purdue University, USA — *Aug 2025*
Major: Aeronautical and Astronautical Engineering

M.S., Purdue University, USA — *May 2020*
Major: Aeronautical and Astronautical Engineering

B.E., Punjab Engineering College (PEC), India — *May 2016*
Major: Materials and Metallurgical Engineering

PROFESSIONAL EXPERIENCE

Assistant Professor (tenure-track) — *Aug 2025-Present*
Aerospace Engineering, Embry-Riddle Aeronautical University

Graduate Researcher — *Jun 2018-Aug 2025*
Space Information Dynamics Group, Purdue University

Graduate Engineer Trainee — *Jun 2016-Jun 2017*
Research and Development, Vardhman Special Steels Limited

Research Intern — *Jan 2015-May 2015*
Bhabha Atomic Research Center

Research Intern — *Jun 2014 - Jul 2014*
Defense Research and Development Organization

RESEARCH GRANTS

Sole PI — *Jul 2026-Jun 2027*
Faculty Innovative Research in Science and Technology (FIRST), ERAU
Title: Optimized Space-Based Observer Architectures for Sustained Cislunar Monitoring

HONORS, AWARDS, RECOGNITIONS

- Estus H. and Vashti L. Magoon Teaching Award, College of Engineering, Purdue University, 2025
- Future Leaders in Aerospace Symposium, (MIT AeroAstro, Stanford, CU Boulder and Penn State), 2024
- Amelia Earhart Leadership for Space Careers Program, 2023-2024
- Trailblazers in Engineering Fellow, Purdue University, 2023
- AAE Teaching Fellowship, School of Aeronautics and Astronautics, Purdue University, 2023-2024
- Prime Minister's Scholarship, Ministry of Defence, India, 2012-2016
- Institute Color (highest extra-curricular activities award), Punjab Engineering College, India, 2016

PUBLICATIONS

Peer-Reviewed Journal Articles

1. J.L. Zapata, E.A. Turner, A. Abdrabou, **S. Bhadauria**, F. Crespo. Secular Attitude Dynamics of an Axisymmetric Rigid Body in a Circular Orbit.
Journal of Astronautical Sciences, April 2026 (In review)

2. **S. Bhadauria**, C. Frueh, Optimal Optical Cislunar Space Surveillance Positioning Leveraging the BCR4BP and Average Visibility.
Advances in Space Research, June 2026
3. **S. Bhadauria**, A. Black, C. Frueh, Cislunar Key Region Surveillance Optimization.
The Journal of Astronautical Sciences, September 2025
4. B. Baker-McEvilly, **S. Bhadauria**, D. C. Garcia, C. Frueh, A Comprehensive Review on Cislunar Operations and Expansion.
Progress of Aerospace Sciences, July 2024

Conference Proceedings

1. S. Quan, **S. Bhadauria**, Optimal Placement of Earth and Moon Ground-Based Observers for Cislunar Space Surveillance.
ASCEND Conference, Washington, D.C., May 2026
2. S. Quan, B. Baker-McEvilly, D. Canales, **S. Bhadauria**. Constellation Design for Cislunar Space Surveillance Based on Optimal Visibility.
AAS/AIAA Astrodynamics Specialist Conference, Whistler, BC, Canada, July 2026 (abstract accepted)
3. L. Bottero, T. A. Lovell, **S. Bhadauria**. Multi-Sensor Data Fusion for Enhanced Cislunar Space Domain Awareness Using Radar and Optical Observations.
AAS/AIAA Astrodynamics Specialist Conference, Whistler, BC, Canada, July 2026 (abstract accepted)
4. A. Abdrabou, F. Crespo, **S. Bhadauria**. Global Error Characterization of Circular Intermediary Model for Attitude-Orbital Dynamics.
AAS/AIAA Astrodynamics Specialist Conference, Whistler, BC, Canada, July 2026 (abstract accepted)
5. **S. Bhadauria**, C. Frueh, Optimal Visibility-Based Design of Surveillance Trajectories in the Cislunar Region.
AAS/AIAA Astrodynamics Specialist Conference, Boston, Massachusetts, August 2025
6. M. Nishiguchi, **S. Bhadauria**, C. Frueh, Characterization of Lunar Consequences of Fragmentation Events Involving Nuclear-Powered Spacecraft in the Cislunar Region.
9th European Conference on Space Debris, April 2025
7. **S. Bhadauria**, A. Black, C. Frueh, Cislunar Surveillance Optimization and Key Region Identification.
Advanced Maui Optical and Space Surveillance Conference, Maui, September 2024
8. B. Baker-McEvilly, **S. Bhadauria**, J. Rose, and D. C. Garcia, C. Frueh, H. Cho, Performance of Observational Spacecraft Across Orbit Families for Space Domain Awareness in the Cislunar Realm.
AIAA SCITECH 2024, Orlando, Florida, January, 2024
9. **S. Bhadauria**, C. Frueh, Optical Observation Regions in Cislunar Space Using Bicircular Four Body Problem Geometry.
Advanced Maui Optical and Space Surveillance Conference, Maui, September 2022
10. **S. Bhadauria**, C. Frueh, K. Howell, Cislunar Space Domain Awareness Using Bicircular Four Body Problem.
AAS/AIAA Astrodynamics Specialist Conference, Charlotte, North Carolina, August 2022
11. C. Frueh, K. DeMars, K. Howell, **S. Bhadauria**, M. Gupta, Cislunar Space Traffic Management: Surveillance Through Earth-Moon Resonance Orbits.
8th European Conference on Space Debris, April 2021 (Virtual conference)
12. C. Frueh, K. Howell, K. DeMars, **S. Bhadauria**, Cislunar Space Situational Awareness.
AAS Space Flight Mechanics Meeting, February 2021 (Virtual conference)

13. **S. Bhadauria**, C. Frueh, Association of Too Short Arcs Using Admissible Regions.
AAS/AIAA Astrodynamics Specialist Conference, August 2020 (Virtual conference)

14. V. Dubey, **S. Bhadauria** and V. Kain, Effect of Dissolved Oxygen on the Corrosion Behavior of Carbon Steel in the Simulated Reactor Environment.
17th Asian Pacific Corrosion Control Conference, Mumbai, January 2016

Thesis and Dissertation

- **S. Bhadauria**, Eyes on the Cislunar Frontier: Enhancing Space Domain Awareness with Ground and Space-Based Observers.
Ph.D. Dissertation, Purdue University, August 2025
- **S. Bhadauria**, Association of Too-Short Arcs Using Admissible Regions.
M.S. Thesis, Purdue University, May 2020

Poster Presentations and Symposia

- S. Quan, **S. Bhadauria**. Constellation Design for Cislunar Space Surveillance Based on Optimal Visibility.
Discovery Day, Embry-Riddle Aeronautical University, April 2026
- L. Bottero, Thomas A. Lovell, **S. Bhadauria**. Multi-Sensor Data Fusion for Enhanced Cislunar Space Domain Awareness Using Radar and Optical Observations.
Discovery Day, Embry-Riddle Aeronautical University, April 2026
- **S. Bhadauria**, Space Observation and Astrodynamics Research (SOAR) Lab Overview.
IAB Research Showcase, Embry-Riddle Aeronautical University, November 2025
- **S. Bhadauria**, Cislunar Space Surveillance.
Future Leaders in Aerospace Symposium, Stanford University, May 2024
Trailblazers in Engineering Poster Session, Purdue University, July 2023
- **S. Bhadauria**, Orbit Determination from Single Observations.
Engineering 2169 Competition, Purdue University, April 2019

Talks and Presentations

- Space Domain Awareness in the Earth-Moon System
Industry Advisory Board, Embry-Riddle Aeronautical University, April 2026
- Cislunar Space Surveillance.
Embry-Riddle Aeronautical University, March 2025
Mississippi State University, February 2026
South Dakota School of Mines, February 2026

TEACHING EXPERIENCE

Instructor *Aug. 2025-May 2026*
Aerospace Engineering, Embry-Riddle Aeronautical University
Spring 2026: Space Mechanics, (AE 313-01), 32 students, Teaching Evaluation: 4.57/5
Spring 2026: Space Mechanics (AE 313-03), 30 students, Teaching Evaluation: 4.40/5
Fall 2025: Space Mechanics (AE 313-01), 40 students, Teaching Evaluation: 3.95/5
Fall 2025: Space Mechanics (AE 313-04), 40 students, Teaching Evaluation: 3.79/5

Instructor of Record *Jan 2024-May 2024*
School of Aeronautics and Astronautics, Purdue University
Spring 2024: Dynamics and Vibrations (AAE 340-03), 57 students, Teaching Evaluation: 4.23/5

Teaching Assistant (Dynamics and Vibrations) *Aug 2023-Dec 2023*
School of Aeronautics and Astronautics, Purdue University

Graduate Assistant (Course Design and Development Team) *Aug 2018-Dec 2022*
Teaching and Learning Technologies, ITaP, Purdue University

Program Assistant *Jun 2018-Jul 2018*
Minority Engineering Program Summer Workshop, Purdue University

Instructor and Course Developer *Jan 2018-Mar 2018*
Gifted Education Research and Resource Institute, Purdue University

MENTORING EXPERIENCE

Ph.D. Dissertation Chair

- Abdulrahman Abdrabou *Jan 2026-present*

M.S. Thesis Chair

- Jacob Blanton *Jan 2026-present*
- Stephanie Quan *Aug 2025-present*
- Lucas Bottero *Aug 2025-present*
- Jayden Dougal *Aug 2025-May 2026*

Ph.D. Dissertation Committee Member

- Brian P. Baker-McEvilly *Aug 2025-present*

M.S. Thesis Committee Member

- Rahi Paragbhai Shah *Jan 2026-present*
- Mauro A. Palomo *Aug 2025-present*

Undergraduate Research

- Stone Lucksinger *May 2026-present*
- Robert Nelson *May 2026-present*
- Joshua Robinett *May 2026-present*
- Joseph Cox *May 2026-present*
- John Lachnicht *May 2026-present*
- Ella Sparks *Jan 2026-present*
- Max Rabbe *Jan 2026-present*
- Giovanni Decapua *Jan 2026-present*
- Ayden Rooks *Jan 2026-May 2026*

SKY-CARE Mentor for Two High School Students *May 2026-present*
ASPIRE, Embry-Riddle Aeronautical University

SKILLS

Programming languages: MATLAB, Python, C++, C
Astrodynamics software: System Tool Kit (STK)
Data Analysis Tools: Origin Pro, Google Data Studio
Creative Suite: Adobe Photoshop, Adobe Premiere Pro, Canva, Camtasia, VideoScribe
Others: Microsoft Suite, Blackboard, edX, Brightspace

SERVICE

Professional Community Reviewer *2023-present*

- Geospatial Information Science
- Journal of Astronautical Sciences
- Advances in Space Research
- Journal of Spacecraft and Rockets
- 2026 AAS/AIAA Astrodynamics Specialist Conference

Department Service
Dynamics and Control Faculty Search Committee Member *Jan 2026-May 2026*

Outreach and Leadership

- Volunteer, Girls in Aviation Day, ERAU *Sep 2025*
- Mentor, AeroAssist, Purdue University *2018-2023*
- Project Coordinator, Asha for Education, Purdue Chapter *Feb 2021-Sep 2022*
- Mentor, College Mentor for Kids, Purdue Chapter *Aug 2021-Dec 2021*
- Event Coordinator, Asha for Education, Purdue Chapter *Aug 2017-Aug 2018*
- Volunteer, NSS, Punjab Engineering College *Aug 2012-Dec 2013*
- General Secretary, Art and Photography Club, PEC *Jul 2015-May 2016*
- Creative Head, PECFEST-15 (Cultural and Technical Fest) *Aug 2015-Nov 2015*

WORKSHOPS ATTENDED

- CISE NSF CAREER Workshop (Virtual), *May 2026*
- Engineering Academic Career Club Mentorship Program, Purdue University *2023*

REFERENCES

- Carolin Frueh
  Harold DeGroff Professor of Aeronautics and Astronautics
  Purdue University
  cfrueh@purdue.edu, (765)-494-7436

- Kathleen Howell
  Hsu Lo Distinguished Professor of Aeronautics and Astronautics
  Purdue University
  howell@purdue.edu, (765)-494-5786

- Kyle DeMars
  Professor of Aeronautics and Astronautics
  Purdue University
  kdemars@purdue.edu, (765)-496-2658

- Kiril Datchev
  Associate Professor
  Department of Mathematics, Purdue University
  kdatchev@purdue.edu